

Is golf a *momentum* sport?

Testing i.i.d. against a hierarchical AR(1) for round-to-round residuals
within a single tournament.

MOTIVATION

Golf is talked about as a *momentum* sport.

“It would be silly to say I don’t ride some of that momentum”

SCOTTIE SCHEFFLER

“We’re going to try and just use last year as momentum”

JORDAN SPIETH

“I feel like my game is trending in the right direction”

JON RAHM

“I’m in a nice rhythm right now”

XANDER SCHAUFFELE

Form has been rigorously studied across tournaments and across seasons. To build a *live, in-tournament* Bayesian updating model for win probability, we need the same answer at the round-to-round level *within a single tournament*.

THE GAP

Form is settled historically. Round-to-round persistence is under examined.

HISTORICAL · DATAGOLF

Which skills predict the next round?

- **What they did:** Use each player's career-to-date average in OTT, APP, ARG, PUTT to predict their next round's total SG.
- **Findings:** Off-the-tee is the most reliable signal ($\beta = 1.2$). Putting is the least ($\beta = 0.6$).
- **Horizon:** Pooled across the player's whole history. A skill profile, not a within-tournament dynamic.

WITHIN-TOURNAMENT · THIS PROJECT

Does Thursday's round predict Friday's, same event?

- **What we test:** Take ε from round r at tournament T . Ask how much shows up in ε at round $r+1$, same T .
- **Target:** ρ_{pop} . Population-average round-to-round persistence after skill and conditions are stripped.

Research question. Is $\rho_{\text{pop}} > 0$ once player skill and round conditions are stripped out?

MEASUREMENT

Strokes Gained, and why we engineer it.

- **What it is.** The expected number of strokes a player gains or loses relative to the field on a given shot, based on the starting and ending positions of the ball.
- **What we have.** Round-level SG by component (OTT, APP, ARG, PUTT) for PGA Tour and LIV events, 2019–2026. ~134K rounds, 674 players. Source: DataGolf.
- **What's wrong with raw SG.** The baseline is field-relative per round. +1 SG against a major field is not +1 SG against a weaker field event. AR(1) on raw SG conflates round-to-round dynamics with cross-tournament field-strength differences.
- **What we do.** Per SG component, fit a Bayesian linear regression with $\mathcal{N}(0, 1)$ priors on player skill (α) and round difficulty (δ). MAP estimate via ridge. Model the residual ε .

METHOD · RESIDUAL CONSTRUCTION

We need round-pair ε 's. Two steps to get them.

$$\text{SG}_{p,T,r} = \alpha_p + \delta_{T,r} + \varepsilon_{p,T,r}$$

- $\varepsilon_{p,T,r}$ — **The residual we need.** SG with player skill and round difficulty removed. Round-pair input to AR(1).
- α_p — **Player skill.** Pre-round expectation for player p , the Monday before the event.
- $\delta_{T,r}$ — **Round difficulty.** Field strength, course, weather, pin, tee draw; absorbed into one indicator.

Step 1 — Bayesian linear regression. $\mathcal{N}(0, 1)$ priors on player skill (α) and round difficulty (δ). MAP estimate via ridge, walk-forward per Monday on the prior 5 years.

Step 2 — Residual. $\varepsilon = \text{SG} - \hat{\alpha} - \hat{\delta}$, with $\hat{\alpha}$ from the snapshot strictly before event date. ε feeds AR(1).

METHOD · WORKED EXAMPLE

One player, one tournament, three pairs.

DECOMPOSITION · SCHEFFLER, 2025 OPEN CHAMPIONSHIP

ROUND	SG_TOTAL	α	δ	ε
R1	+5.04	+3.16	-0.42	+2.29
R2	+7.41	+3.16	-0.23	+4.47
R3	+2.97	+3.16	-0.75	+0.56
R4	+1.89	+3.16	-0.79	-0.49

AR(1) INPUT PAIRS

PAIR	ε_r		ε_{r+1}
R1 → R2	+2.29	→	+4.47
R2 → R3	+4.47	→	+0.56
R3 → R4	+0.56	→	-0.49

$\varepsilon = \text{sg_}^* - \alpha - \delta$. Four rounds yield three consecutive pairs.

MODELS

Independence vs. hierarchical AR(1). PSIS-LOO decides.

MODEL A · INDEPENDENCE

Each round residual is an independent draw.

LIKELIHOOD $y_{p,T,r} \sim \mathcal{N}(0, \sigma^2)$

PRIOR $\sigma \sim \text{Half-}\mathcal{N}(2)$

$y_{p,T,r}$ Round residual for player p , tournament T , round r .
 σ Round-level noise scale.

MODEL B · HIERARCHICAL AR(1)

Round r is a partial echo of round $r-1$.

LIKELIHOOD $y_{p,T,r} \mid y_{p,T,r-1} \sim \mathcal{N}(\rho_p y_{p,T,r-1}, \sigma^2)$

PRIORS $\rho_p \sim \mathcal{N}(\rho_{\text{pop}}, \tau_\rho^2)$
 $\rho_{\text{pop}} \sim \mathcal{N}(0, 0.5^2)$
 $\tau_\rho \sim \text{Half-}\mathcal{N}(0.3), \sigma \sim \text{Half-}\mathcal{N}(2)$

ρ_p Player p 's round-to-round persistence.
 ρ_{pop} Tour-wide average persistence: our research target.
 τ_ρ Between-player SD. Partial pooling: small τ_ρ shrinks ρ_p toward ρ_{pop} , large τ_ρ lets players deviate.

Decision rule. PSIS-LOO: if Model B beats Model A by >2 SE on out-of-sample log density, the AR(1) signal is real.

RESULTS · FIRST PASS

AR(1) wins on every component. Estimate is suspect.

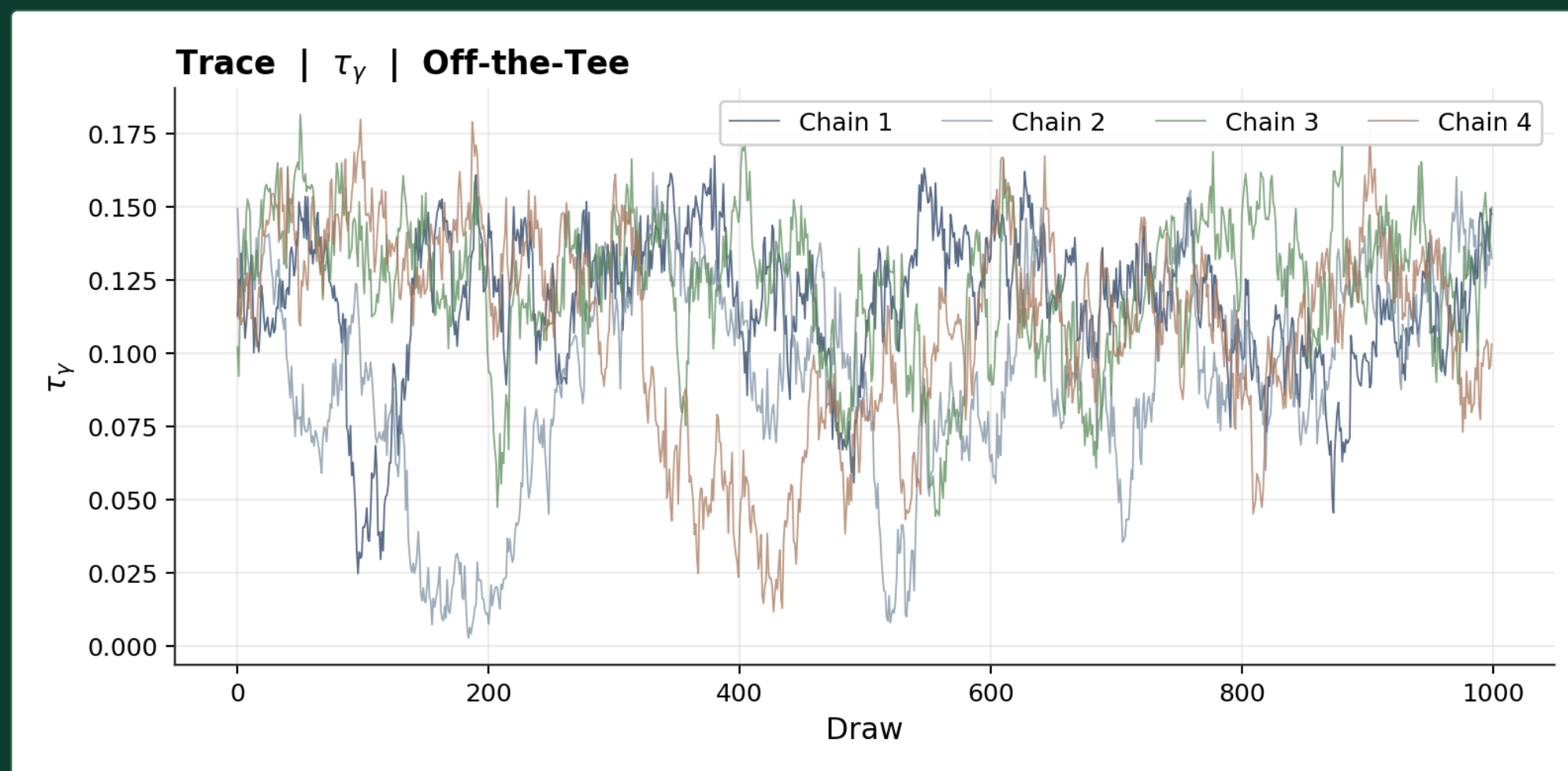
COMPONENT	ρ_{pop} MEAN	95% CI	LOO Δ	Δ / SE
Off-the-Tee	0.086	[0.077, 0.095]	249	9.1
Total	0.045	[0.038, 0.052]	88	6.2
Approach	0.033	[0.026, 0.041]	35	4.0
Around-the-Green	0.030	[0.022, 0.037]	27	3.3
Putting	0.020	[0.014, 0.029]	13	2.4

The issue. Any constant offset across a player’s four rounds at one tournament (course fit, week conditions, stale skill snapshot) mechanically inflates ρ_{pop} . Adding random effects to control for this allows for a more robust test of whether momentum is the driving factor.

METHOD · THE PROBLEM

Splitting course fit from momentum: Add random effects.

- Course fit, week conditions, and stale skill snapshots shift all four of a player's rounds by the same amount. AR(1) reads that shared shift as momentum.
- The solution: add random effects $\gamma_{p,T}$. One number per player-tournament, absorbing the shared shift.
- However, sampling 33,000+ $\gamma_{p,T}$ directly breaks convergence.



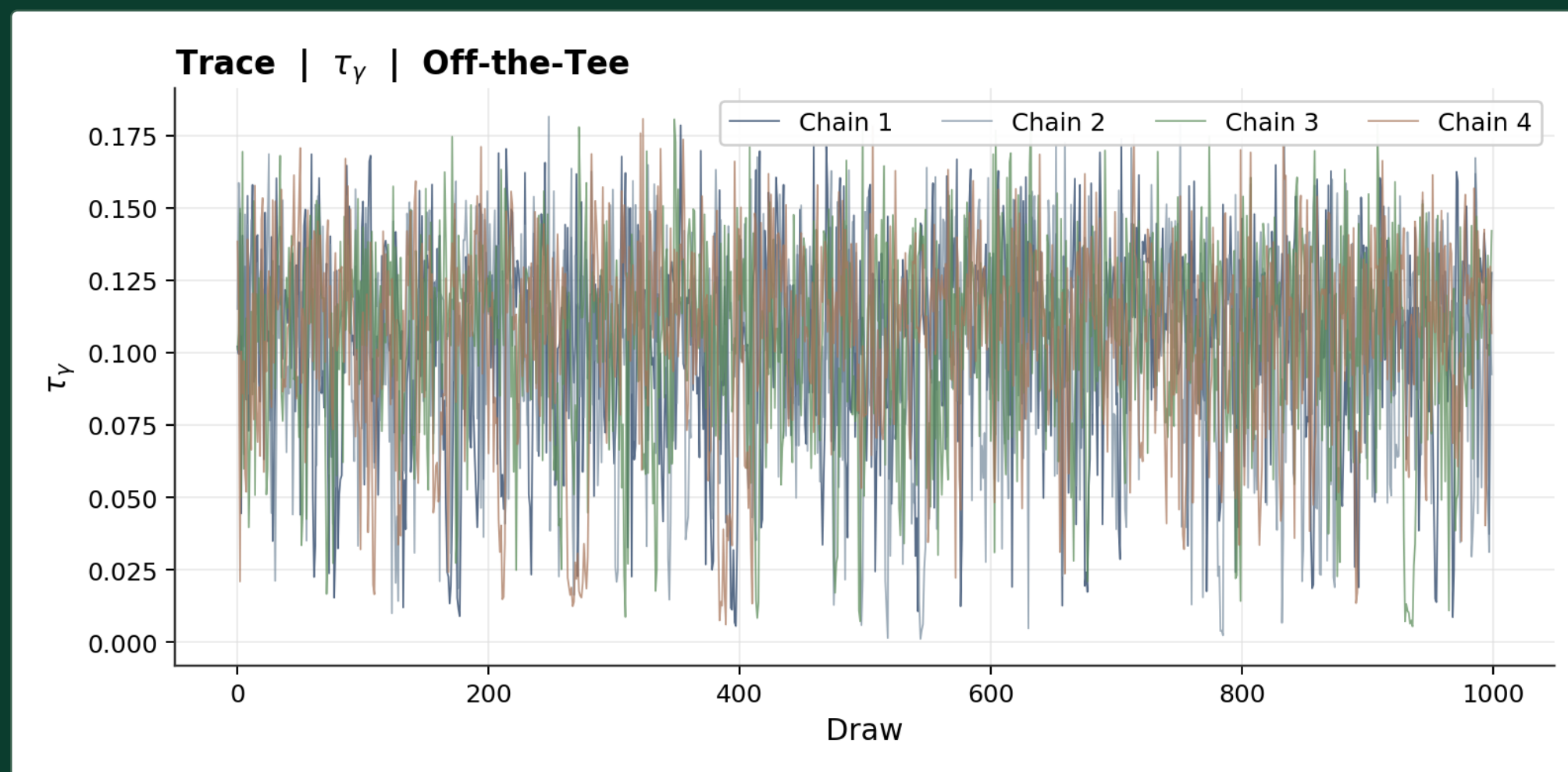
Two chains collapse near zero. Funnel geometry, $\hat{R} = 1.08$, ESS = 37.

Implementation problem. 33k non-centered latents. Tightening τ_γ from Half- $\mathcal{N}(0.5)$ to Half- $\mathcal{N}(0.2)$ traded poor mixing for 17% divergent transitions. The funnel is geometric.

METHOD · THE FIX

Marginalize γ . Convergence restored.

- Since the sum of two Gaussians is Gaussian, γ folds analytically into the likelihood's covariance.
- We get identical posteriors on $(\rho_{\text{pop}}, \tau_\rho, \tau_\gamma, \sigma)$, with zero divergences and an $\hat{R}$ that drops below 1.01.
- The marginalized form with all 33,000+ $\gamma_{p,T}$ integrated out is what we then sample.



Four chains mix around the same mean. Healthy posterior, $\hat{R} = 1.00$.

What we lose. Per-player-tournament γ values. The control on ρ is identical; we just don't print them.

METHOD · MARGINALIZATION

Same model. γ integrated out.

LIKELIHOOD

$$y_{p,T,r} \mid y_{p,T,r-1}, \gamma_{p,T} \sim \mathcal{N}(\rho_p y_{p,T,r-1} + \gamma_{p,T}, \sigma^2)$$

 γ PRIOR

$$\gamma_{p,T} \sim \mathcal{N}(0, \tau_\gamma^2)$$

AFTER
MARGINALIZING γ

$$\mathbf{y}_c \sim \text{MVN}(\boldsymbol{\mu}_c(\rho_p), \sigma^2 \mathbf{I}_k + \tau_\gamma^2 \mathbf{J}_k)$$

PRIORS

$$\begin{aligned} \rho_p &\sim \mathcal{N}(\rho_{\text{pop}}, \tau_\rho^2), & \rho_{\text{pop}} &\sim \mathcal{N}(0, 0.5^2) \\ \tau_\rho &\sim \text{Half-}\mathcal{N}(0.3), & \tau_\gamma &\sim \text{Half-}\mathcal{N}(0.2), & \sigma &\sim \text{Half-}\mathcal{N}(2) \end{aligned}$$

Note. Marginalized form is what we sample. γ posteriors are not retained.

RESULTS · WITH RANDOM EFFECT

ρ_{pop} barely moves. The signal survives.

COMPONENT	ρ_{pop} MEAN	VS. FIRST PASS	95% CI	LOO Δ	Δ / SE
Off-the-Tee	0.080	−7%	[0.069, 0.091]	124	5.5
Total	0.045	0%	[0.037, 0.052]	76	6.3
Approach	0.031	−6%	[0.023, 0.040]	20	3.0
Around-the-Green	0.027	−10%	[0.019, 0.036]	12	2.0
Putting	0.018	−10%	[0.009, 0.027]	5	1.2

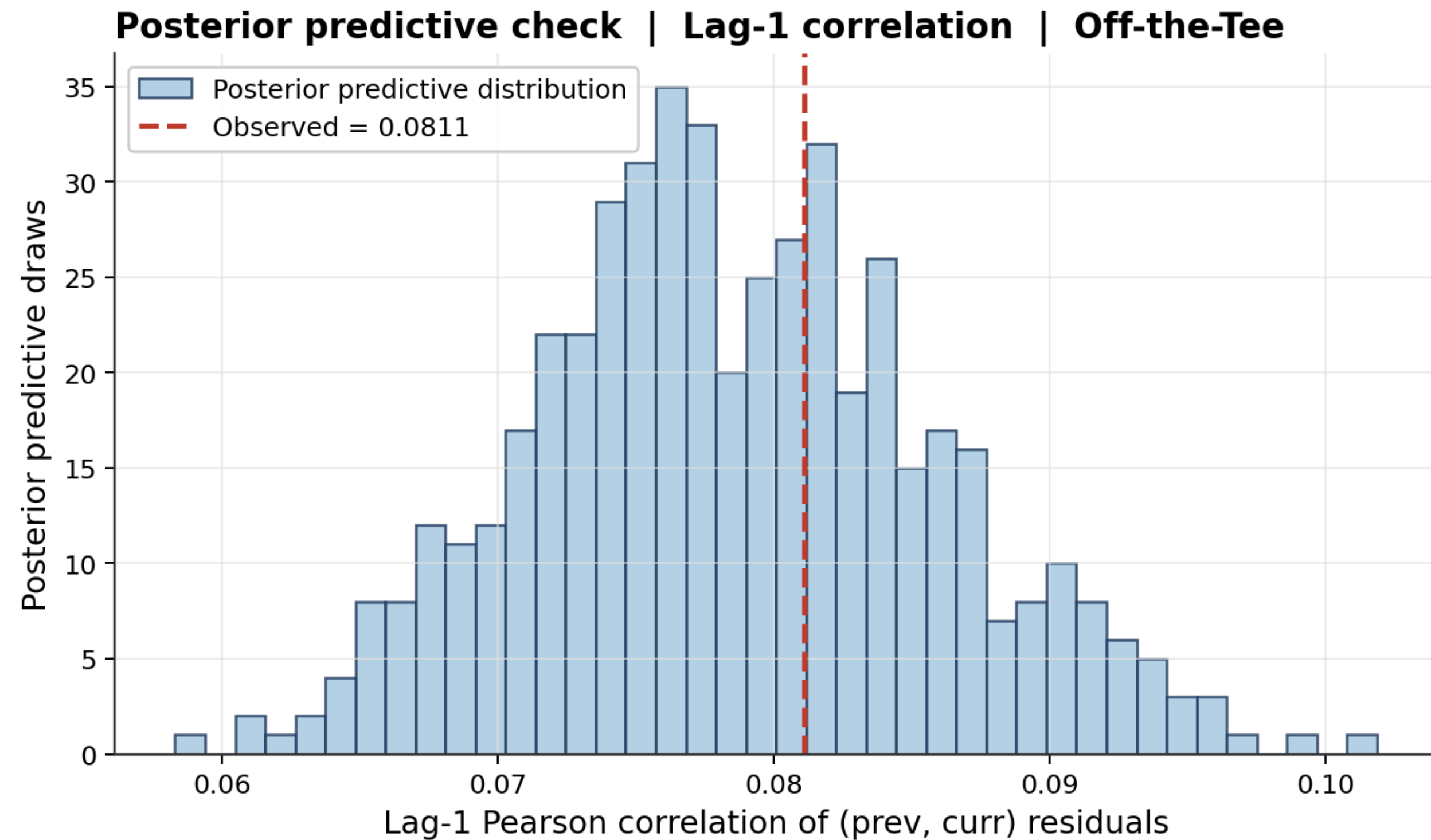
Read. Reduction is at most 10%. CIs widen slightly, which is the honest cost of the control. Component ranking is preserved. The AR(1) signal is not an artifact of the course fit offset.

Prior sensitivity (OTT). ρ_{pop} unchanged ($\Delta < 0.001$) under ρ_{pop} priors $\mathcal{N}(0, 0.2^2)$ – $\mathcal{N}(0, 1.0^2)$ and τ_γ priors Half- $\mathcal{N}(0.2)$ –Half- $\mathcal{N}(0.5)$. Data dominates.

VALIDATION · POSTERIOR PREDICTIVE CHECK

The model reproduces the lag-1 correlation it claims.

LAG-1 CORRELATION · OFF-THE-TEE



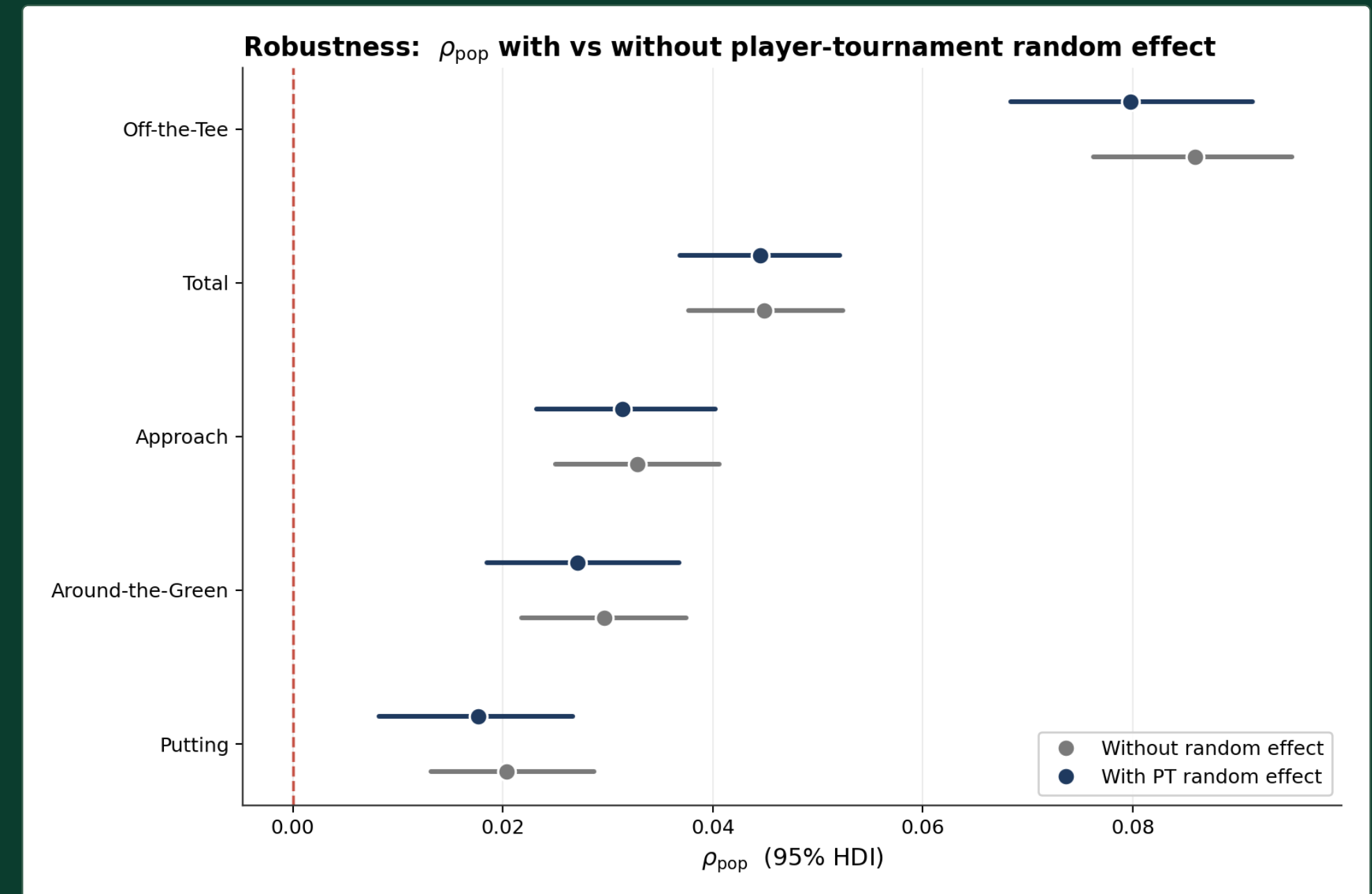
500 posterior predictive draws. Observed value falls in the bulk of the simulated distribution.

Result. Observed lag-1 correlation 0.081. Simulated mean 0.079 ± 0.007 ($p = 0.72$). The AR(1) structure reproduces the headline statistic.

FINDINGS

AR(1) is justified. Persistence decays toward the hole.

- **AR(1) is the right framework.** PSIS-LOO prefers Model B over independence on four of five components ($\Delta/\text{SE} > 2$). The iid null is rejected, and we confirm there is some momentum between rounds.
- **Persistence decays toward the hole.** OTT (0.080) > APP (0.031) > ARG (0.027) > PUTT (0.018). Monotone ordering: the further from the hole, the more round-to-round signal carries. Ordinal rankings of components follow DataGolf's historical-to-next-round result.
- **Robust to random effects.** Marginalized random effect decreases ρ_{pop} by $\leq 10\%$ across components. Course fit, week conditions, and snapshot error don't explain it away.



Verdict. A Markov structure is justified for the live Bayesian win-probability model.

STRENGTHS & WEAKNESSES

What this analysis does well, and what it doesn't.

STRENGTHS

What the analysis does well.

- **lid null rejected.** PSIS-LOO prefers AR(1) on four of five components. $\rho_{\text{pop}} > 0$ with non-overlapping CIs.
- **Component ordering reproduces DataGolf.** OTT > APP > ARG > PUTT, within-tournament signal matches their across-career result.
- **Cluster shift controlled for.** Marginalized γ absorbs course fit, week conditions, snapshot staleness. ρ_{pop} drops $\leq 10\%$ with the control in.

WEAKNESSES

What the analysis doesn't address.

- **Pre-computed inputs.** α and δ are ridge point estimates. Uncertainty in those inputs doesn't propagate into the posterior on ρ_{pop} .
- **Gaussian likelihood underestimates tails.** Posterior predictive check shows $\sim 3\times$ more $|\varepsilon| > 3$ events than the model predicts. A Student- t likelihood would absorb extreme rounds. ρ_{pop} estimate is unaffected.

REFERENCES

References.

Methodology and prior work cited in this analysis.

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